

A Low-Power Bluetooth LE Surface EMG Sensor

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Abstract— We present a low-power Bluetooth Low Energy (BLE) surface electromyography (sEMG) system designed for wearable muscle monitoring. The device integrates a fully custom signal acquisition pipeline — including analog filtering, signal conditioning, dataset-driven waveform reproduction, and BLE-based wireless transmission — centered around a $\Delta\Sigma$ 16-bit ADC and a Nordic nRF52840 SoC. To evaluate its performance, a representative sidekicking motion from a public sEMG dataset was physically reproduced and recorded by both the proposed system and a clinical-grade EMG device. Time and frequency domain analyses show that the system preserves the key features of myoelectric signals within the 30–400 Hz band. Results support its suitability for applications in sports science, rehabilitation, and human–machine interaction.

Keywords — Wearable EMG, low-power, Bluetooth Low Energy, signal-to-noise ratio, sports monitoring.

I. INTRODUCTION

Surface electromyography (sEMG) records the extracellular potentials generated by motor-unit action, offering a direct, non-invasive proxy of neuromuscular drive that can be monitored in real time [1, 2]. Because sEMG amplitude scales with the number of recruited fibres and its spectral content reflects conduction velocity, it has become essential in rehabilitation, ergonomics, sports science and human–machine interfaces [3, 4]. Clinicians exploit sEMG for biofeedback after stroke, engineers embed it in myoelectric prostheses and exosuits, and coaches track muscle-specific fatigue to refine training loads.

Commercial systems, however, remain largely confined to the laboratory. They rely on wired electrodes, bulky front-ends and high-power radios that restrict sessions to a few hours and tether the subject to fixed equipment [5]. These limitations hinder long-term studies of fatigue, out-of-clinic physiotherapy and field deployments where comfort, discretion and zero-maintenance are paramount.

The convergence of low-noise instrumentation amplifiers, high-resolution $\Delta\Sigma$ ADCs and Bluetooth Low Energy (BLE) SoCs now enables genuinely wearable EMG. Prototypes span dry-electrode armbands [3], textile arrays woven into garments [6], milliwatt EEG/EMG headsets with on-device compression [7], and month-scale BLE loggers [4]. Yet

meeting clinical-grade signal-to-noise ratio (SNR) under coin-cell power still demands careful co-design of analogue front-end, firmware and radio scheduling. Noise, quantisation and packet overhead remain the dominant bottlenecks.

In this context, we propose a low-power sEMG node that fits on a 55 mm $\times$ 30 mm PCB and streams one 16-bit channel at 2 kS/s over BLE. Key design choices include a programmable-gain 16-bit ADS112C04 optimized for the 30–400 Hz myoelectric band, and a firmware pipeline that buffers 1 kS/s data into 20 Hz BLE notifications, significantly reducing on-air time without sacrificing resolution. Performance was evaluated by physically reproducing a representative sidekicking motion from a public sEMG dataset and comparing the acquired signals to those recorded simultaneously with a clinical-grade EMG system. Results show that even with milliwatt operation, the proposed system achieves high signal fidelity and reduced motion artifacts.

The remainder of this paper is organized as follows: Section II presents the hardware and firmware design of the proposed sEMG system; Section III describes the data acquisition protocol based on physical reproduction of a public dataset segment; Section IV reports validation results through comparative analysis with a clinical EMG system; and Section V concludes the paper and discusses future work.

II. MATERIALS AND METHODS

A. System overview

Since sEMG signals represent the summed electrical activity of active motor units (MUs) detected by electrodes placed on the skin surface above the muscle, they are typically characterized by low amplitude and low frequency. The signal amplitude usually ranges from 0.01 to 5 mV, with peak-to-peak values below 6 mV and root mean square (RMS) values under 1.5 mV [8]. In terms of frequency, sEMG signals span from 0 and 500 Hz, with the majority of the energy concentrated between 50 and 150 Hz, making it highly susceptible to noise interference [9, 10].

Therefore, the processing circuit must meet requirements such as high input impedance, high gain, high Common Mode Rejection Ratio (CMRR), and low noise [11, 12].

Traditionally, most studies have used wired transmission for sEMG acquisition [13, 14], which can restrict the user's range of motion and hinder remote rehabilitation monitoring. Wireless acquisition, on the other hand, offers greater freedom of movement and enables dynamic remote monitoring.

Figure 1 shows the block diagram of the sEMG signal acquisition circuit. The circuit is divided into seven parts: pre-amplification, high-pass filter (HPF), low-pass filter (LPF), adjustable-gain amplifier, analog-to-digital conversion, reference voltage circuit, and microcontroller.

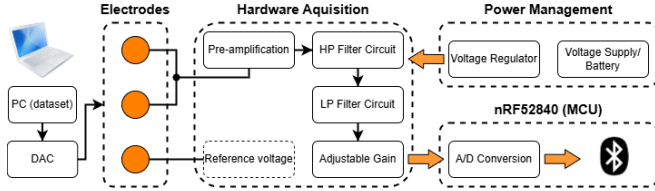


Fig. 1. sEMG acquisition block diagram

The sensor node prototype was assembled on a compact printed circuit board (PCB) populated with surface-mount devices (SMD), as shown in Figure 2. Key functional blocks include the analog front-end (AFE), the microcontroller unit (MCU), High-pass and Low-pass filter circuits, an adjustable-gain integrated circuit (IC), and the ADS114C04 analog-to-digital converter (ADC).

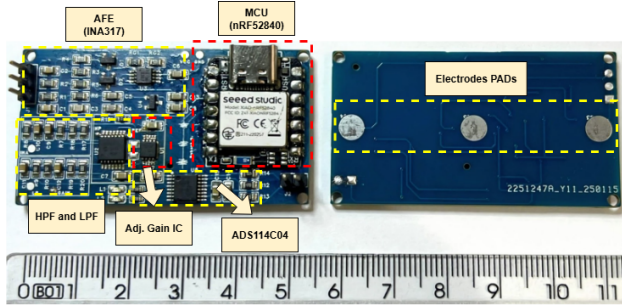


Fig. 2. Images of the fabricated device

Table 1 below presents some characteristics of the designed prototype. The cascaded circuit was designed to have a total gain of 907 and a bandwidth ranging from 30 Hz to 400 Hz.

TABLE 1. Characteristics of proposed sEMG sensor

Description	Value
Number of Channels	1
Sampling Frequency (f_s)	2 KS/s
A/D Resolution (α)	16 bits
Bandwidth (BW)	30Hz - 400Hz
Adjustable Gain (G)	1x - 11x
Communication Type	BLE 5.1
Receiver Type	Computer, Smartphone
Voltage Supply (V_{cc})	5 V
Power Supply	400 mAh Li Battery
Dimensions (mm^2)	55 mm x 30 mm
Electrode Material	Ag/AgCl

B. Electrodes

The prototype was developed with a focus on miniaturization to meet the requirements of portable systems. It was designed to operate with three electrodes: two placed at the ends to perform differential measurement of the sEMG signal, and a third, positioned between them, used as the reference electrode. For signal acquisition, differential Ag/AgCl conductive gel electrodes were used, with a spacing of 25 mm between the active terminals. This differential configuration contributes to the effective suppression of common-mode interference [15].

C. Instrumentation Amplifier

In the pre-amplification circuit (Figure 3), the instrumentation amplifier INA317 was used. The amplifier was chosen due to its low current consumption (50 μ A), low offset voltage (75 μ V), high CMRR (110 dB), and low noise—essential characteristics for measuring low-amplitude signals [16].

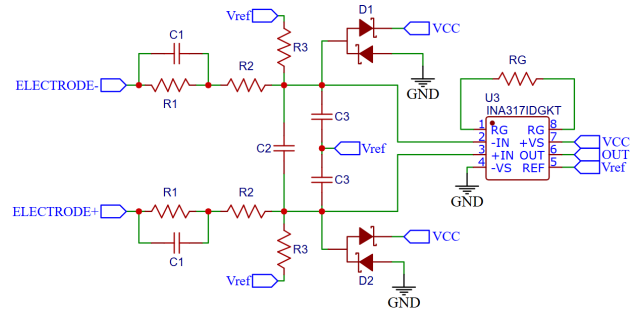


Fig. 3. Schematic of sEMG instrumentation amplifier

Additionally, in the pre-amplification stage, circuits were added for protection against electromagnetic interference (EMI), electrostatic discharge (ESD), overcurrent, and overvoltage [17, 18]. Both negative and positive inputs were connected to resistors (R_1) for overcurrent protection. Protection against EMI and ESD is provided by R_2 , C_2 , and C_3 . BAT54S diodes protect against undervoltage and overvoltage conditions. After an initial low-pass filtering stage, the signal is amplified by the instrumentation amplifier with a fixed gain of 35x.

At this stage, the fixed gain sets the lower limit of the desired range, while higher gains are achieved through the adjustable-gain amplifier circuit.

The V_{ref} voltage injects a small offset current through the R_3 resistors to help the system recover after a stimulation pulse, thus preventing the internal nodes from remaining at excessively high impedance [17].

D. The band-pass filter (HPF and LPF)

Active filters were used in the system design, offering several advantages over passive filters. Among these are impedance isolation between consecutive stages, the ability to function as signal amplifiers, and the use of capacitors as reactive elements instead of inductors, whose precision and accuracy are more difficult to achieve [19].

The filter design process begins with defining the desired

quality factor and cutoff frequency. It is implemented using the Sallen-Key topology with a Butterworth response, ensuring a maximally flat passband [20].

Specifically, a second-order Sallen-Key high-pass filter was designed with a Butterworth response, a cutoff frequency of 20 Hz (measured 30 Hz), and a gain of 1.55. Additionally, a second-order Sallen-Key low-pass filter was implemented with a Butterworth response, a cutoff frequency of 482 Hz (measured 400 Hz), and the same gain of 1.55. The resulting frequency response of the band-pass configuration is illustrated in Figure 4.

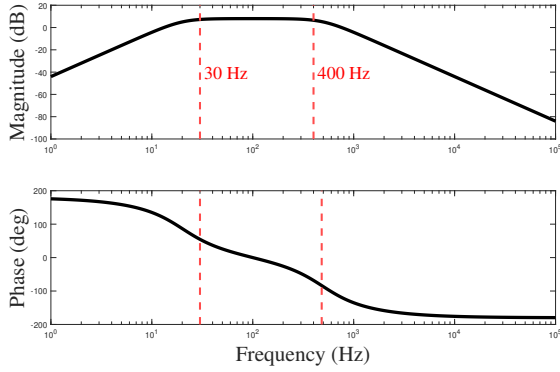


Fig. 4. Bode diagram of the Butterworth band-pass filter

E. The adjustable gain amplifier

The adjustable-gain amplifier circuit is designed to adapt the EMG signal amplitude according to the measurement site, allowing signals from different muscles to be adjusted to ranges compatible with the acquisition system.

The circuit uses the DS3502 IC, a 7-bit digital potentiometer controlled via I²C communication, with a maximum resistance set to 10 k Ω . This configuration enables the acquisition circuit gain to be adjusted within a range from 2x to 11x.

F. Analog-to-Digital Converters

The ADS112C04 is a 16-bit low-power ADC with an I²C interface, ideal for capturing low-amplitude signals. It features a programmable gain amplifier (1x – 128x), four differential channels, adjustable sampling rates up to 2 KS/s, and power-efficient modes (typically 315 μ A). Operating from 2.3 V to 5.5 V, it offers high resolution and flexibility for portable biosensing applications [21]. Its selection was primarily motivated by its low current consumption, high resolution, and configurable gain, making it well-suited for compact and energy-efficient sEMG acquisition systems.

G. Level shifters

sEMG waveforms often swing below 0 V, risking disruption or damage to microprocessor inputs. To prevent this, we designed a DC-offset stage—using a precision 2.048 V reference (MAX6106) and an MCP609 op-amp in an inverting topology—that shifts the signal cleanly into the 0–5 V range required by our processor, with no loss of fidelity.

H. The nRF52840 microcontroller

The nRF52840 Nordic Semiconductor System on Chip (SoC) (Nordic Semiconductor) integrates a 64 MHz Arm Cortex M4F core, 1 MB of on-chip flash, 256 kB of RAM, and a 2.4 GHz multiprotocol radio that supports Bluetooth 5.4 (LE, Long Range). The device was selected for its low-power operation. In our design, the SoC is powered by a single-cell 400mA @ 3.7 V lithium-ion battery; a high-efficiency step-up converter and a low-noise voltage regulator generate the required supply rails for both the analog front-end and the microcontroller.

III. Data Acquisition Protocol

The dataset used in this study was derived from an experimental protocol designed to analyze both normal and aggressive physical movements. Four volunteers (three males and one female, aged 25–30), all with prior exposure to physical aggression scenarios, participated in 20 sessions each, performing ten normal and ten aggressive actions. EMG signals were acquired using a Delsys system with eight surface electrodes placed on the arms (biceps/triceps) and legs (quadriceps/hamstrings). Data were collected in the Essex Robotic Arena (4 $\times$ 5.5 m), where subjects interacted with a 1.75 m punching bag, enabling the acquisition of high-intensity, high-variability signals under realistic conditions. Each session produced eight EMG channels with 10,000 samples, providing a rich basis for evaluating acquisition systems [22].

For validation of the proposed wireless EMG system, a representative sidekicking motion from the dataset was physically reproduced and recorded simultaneously by both the prototype and a clinical-grade EMG system. This enabled direct comparison with the reference signal and assessment of the system's ability to capture rapid, high-amplitude muscle contractions.

A unified signal processing pipeline was applied to both recordings. Signals were first normalized to -1,1, then envelope detection was performed using RMS values over 50 ms sliding windows. Active segments were identified when the envelope exceeded 20% of the signals peak amplitude. Noise estimation used RMS from resting segments, and the signal-to-noise ratio SNR was computed as [5]:

$$\text{SNR}_{\text{dB}} = 20 \log_{10} \left(\frac{\text{RMS}_{\text{signal}}}{\text{RMS}_{\text{noise}}} \right) \quad (1)$$

This approach ensured a fair, quantitative comparison between the two systems, supporting a robust evaluation of the prototype's performance against a clinical reference.

IV. MEASUREMENT RESULTS

A. Time-Domain Performance

To evaluate the performance of the proposed system, a representative sidekicking motion from the dataset was physically reproduced and recorded by both the wireless sEMG prototype and a clinical-grade EMG system. The same filtering, normalization, contraction detection, and SNR

computation (Eq. 1) were applied to both devices, allowing a direct and consistent comparison of signal fidelity and system performance.

Figure 5 shows the normalized electromyographic signals in the time domain, along with their respective envelope curves (in black), for three distinct sources: the original signal from the dataset (top), the signal acquired by the prototype sEMG system (middle), and the signal captured by the clinical EMG system (bottom). The gray areas indicate the segments of muscle contraction detected automatically.

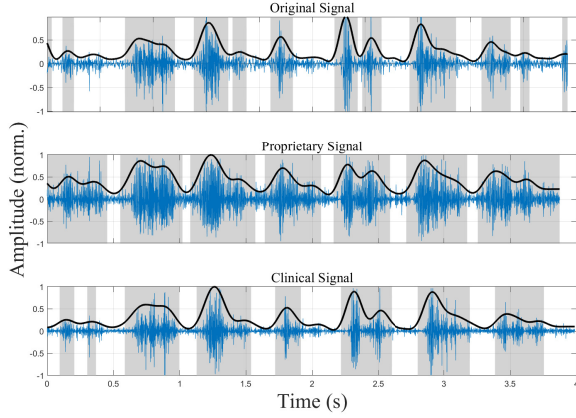


Fig. 5. Normalized EMG signals in the time domain for an aggressive sidekicking movement

By physically reproducing an aggressive sidekicking movement under controlled conditions, it was possible to record the same gesture using both the proprietary and clinical EMG systems, enabling direct comparison with the original dataset. This visualization supports a qualitative assessment of each system's fidelity in capturing muscular dynamics.

The three signals show strong temporal alignment, with synchronized activation peaks, indicating that the muscular event was consistently reproduced. Both systems preserved the overall waveform structure and contraction pattern, as evidenced by the envelope curves. While the proprietary system exhibited slightly more high-frequency noise, it still maintained well-defined peaks and morphological consistency.

The automatically detected contraction regions were similar across all signals, suggesting robustness in the segmentation approach. Signal-to-noise ratio (SNR) analysis yielded values of 10.7 dB for the original dataset, 9 dB for the proprietary system, and 12.7 dB for the clinical system. Although the proposed system presented greater background variability, it did not significantly affect the identification of relevant events.

Overall, these findings demonstrate that the developed wireless EMG system is capable of reliably capturing myoelectric activity during dynamic and intense movements, with performance comparable to clinical-grade solutions in terms of timing, morphology, and muscle contraction detection.

B. Frequency-Domain Performance

Figure 6 shows the frequency response of both evaluated systems: the clinical EMG system (blue) and the proposed prototype (black). Within the physiologically relevant range for surface EMG signals, both systems demonstrate consistent gain, indicating adequate sensitivity to meaningful muscle activity. However, the proposed system exhibits a generally higher gain across most of the spectrum, suggesting enhanced sensitivity—though this may also increase vulnerability to noise and signal saturation.

Regarding phase response, the clinical system shows a more linear and gradual phase shift, which is desirable for preserving temporal integrity across frequency components. In contrast, the proprietary system exhibits sharper phase variations, particularly at higher frequencies, indicating non-uniform group delay that could impair precise timing of rapid events such as contraction onset.

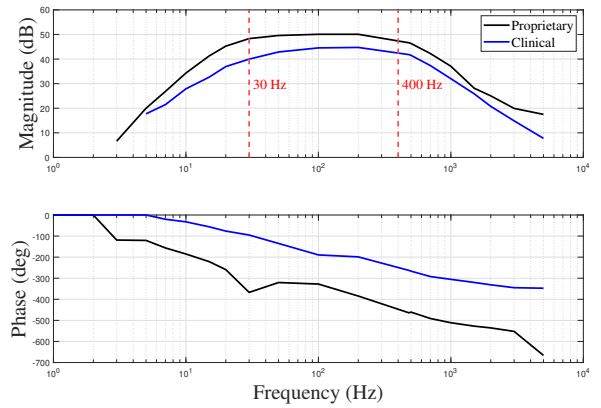


Fig. 6. Frequency response of the proprietary and clinical EMG systems

The clinical system, used as a reference, exhibits a response consistent with what is expected for established medical applications. The comparison shows that the proprietary system is also capable of capturing this type of signal with good quality, preserving its main spectral characteristics. The differences observed in the gain and phase curves reflect design-specific characteristics but do not compromise the proposed system's ability to adequately represent the dynamics of the EMG signal in real-world usage scenarios.

C. Power Consumption

The device works in four distinct power modes: OFF – the entire system is powered down, so both current and power drop to 0; IDLE – electronics are energised, but the node is not yet connected to a central BLE device; CONNECTED – the node is already paired to a smartphone, although the BLE characteristic is *not* being read; TRANSMITTING – the smartphone actively reads the characteristic, so packets are sent periodically.

Table 2 lists the RMS current and the resulting power for each state with a 5.0 V rail; the same windows are highlighted in Fig. 7. The colours are: OFF (purple), IDLE (light-blue), CONNECTED (orange) and TRANSMITTING (pink).

TABLE 3. Comparisons of the sEMG Signal Detection Systems

Systems	Channels	ADC resolution (bits)	Data transmission	Bandwidth (Hz)	Sampling rate (KS/s)
This work	1	16	BLE	30 - 400	2.0
Imperatori et.al, 2013 [23]	4	12	BLE	8.7 - 952	4.0
Tang et.al, 2012 [24]	6	10	RF	20 - 500	1.0
Cerone et.al, 2019 [25]	32	16	Wi-Fi	10 - 500	2.0
Ahamed et.al, 2015 [26]	6	10	Bluetooth	0.03 - 564	1.2
Chen et.al, 2017 [8]	8	12	Wired	20 - 450	1.0

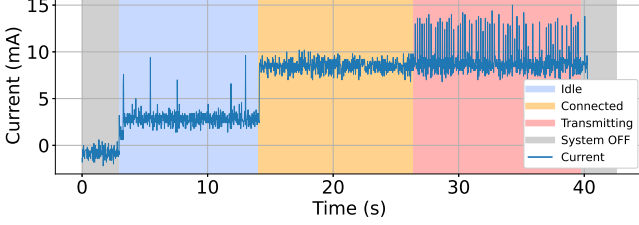


Fig. 7. Current trace with shaded regions for each operating state

The device draws only ~ 2.9 mA (14.6 mW) while **IDLE**. Establishing a connection raises the current in a single step to ~ 8.5 mA. Once streaming begins, periodic current spikes appear, pushing the RMS value to ~ 9 mA and clearly revealing the packet transfers.

Power consumption was minimized at the firmware level through several strategies. All unused peripherals are disabled whenever the radio is disconnected, and Nordic's built-in power management routine is invoked during idle periods to reduce energy consumption. Additionally, the BLE transmission power was lowered to -20 dBm, and the advertising interval was extended from the default 40 ms to 500 ms, further reducing the radio duty cycle and overall power usage.

TABLE 2. RMS current and power for each operating state (5.0V supply)

Window	Current(mA)	Power(mW)
IDLE	2.922	14.61
CONNECTED	8.497	42.49
TRANSMITTING	9.063	45.32
OFF (0V)	0	0

TABLE 4. Comparison of consumption of sEMG hardware architectures

Architecture	Consumption
This Work	45.32mW
Clinical device [27]	65mW
Chen et.al, 2020 [28]	62.7mW
Simić et. al, 2024 [29]	67.65mW

V. CONCLUSION

This work demonstrated that the developed wireless EMG system is capable of reliably capturing myoelectric signals during intense movements, such as sidekicking, using a physically reproduced segment from a public dataset. The temporal response showed good alignment with the reference signal, and the SNR analysis yielded values of 9 dB (proposed system), 12.7 dB (clinical), and 10.7 dB (original), indicating

performance comparable to established solutions.

The frequency response confirmed adequate coverage of the physiological EMG band (30–400 Hz), and the combination of a dedicated analog front-end, a 16-bit $\Delta\Sigma$ ADC, and the nRF52840 SoC enabled clinical-grade signal quality in a compact, low-power device. The node streams data at 2 kS/s consuming only 45 mW, approximately 30% lower than similar systems reported in the literature, as shown in Table 4.

Although the system is limited to a single channel, it delivers comparable performance in terms of resolution, bandwidth, and sampling rate (Table 3). Future work will focus on expanding to multi-channel acquisition and conducting tests with human volunteers, following approval from an ethics committee, to establish the system as a robust solution for use beyond clinical settings.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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